

24CS31 — Worked Solutions

Every question on your 3-hour plan, solved in full. Read the method, then cover the page and reproduce it. Each answer below has been checked numerically.

LAPLACE TRANSFORMS YOU MUST KNOW BY HEART

$f(t)$	$F(s) = L\{f\}$	Note
1	$1/s$	
t^n	$n! / s^{n+1}$	$t \rightarrow 1/s^2, t^2 \rightarrow 2/s^3$
e^{at}	$1/(s-a)$	
$\sin at$	$a/(s^2+a^2)$	
$\cos at$	$s/(s^2+a^2)$	
$\sinh at$	$a/(s^2-a^2)$	
$\cosh at$	$s/(s^2-a^2)$	

THE PROPERTIES — THESE ARE WHAT THE QUESTIONS ACTUALLY TEST

Name	Rule	When you use it
Definition	$L\{f\} = \int_0^\infty e^{-st} f(t) dt$	every proof starts here
First shifting	$L\{e^{at} f(t)\} = F(s-a)$	an e^{at} multiplying anything
Second shifting	$L\{g(t-a)u(t-a)\} = e^{-as}G(s)$	Heaviside / piecewise questions
Multiply by t^n	$L\{t^n f\} = (-1)^n \frac{d^n F}{ds^n}$	Q1c proof, and $L\{t \sin at\}$ type
Divide by t	$L\{f/t\} = \int_s^\infty F(u) du$	anything over t
Derivative	$L\{y'\} = sY - y(0)$ $L\{y''\} = s^2 Y - s y(0) - y'(0)$	every ODE question
Integral	$L\{\int_0^t f(u) du\} = F(s)/s$	an integral inside the transform
Periodic	$L\{f\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$	period- T functions
Inverse of F	$L^{-1}\{F(s)\} = -t f(t)$	inverse of a log or $\tan^{-1}$

LINEAR ALGEBRA FORMULAS

Name	Rule	Where
Linear map test	$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and $T(k\mathbf{u}) = kT(\mathbf{u})$	Q5b
Rank-nullity	rank + nullity = number of columns	Q6c, Q7
Normal equations	$A^T A \mathbf{x} = A^T \mathbf{b}$	least squares, Q8b
Orthogonal diagonalization	$P^T A P = D$, columns of P = unit eigenvectors	Q10c
SVD	$A = U \Sigma V^T$, $\sigma_i = \sqrt{\lambda_i}$ of $A^T A$, $\mathbf{u}_i = A \mathbf{v}_i / \sigma_i$	Q9c
Conjugate transpose	A^* = transpose, then flip the sign of every i	Q10a, Q10b
Hermitian / Unitary	$A^* = A$ / $A^* A = I$	Q10b
Positive definite	ALL leading principal minors > 0 (Sylvester)	Q10a
Rotation matrix	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, reflect in x: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Q5c, Q6a

1. Unitary and Hermitian check

Unit V · Q10b · 5 marks · 8 minutes to learn

THE QUESTION

(a) Determine whether $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ is unitary.

(b) Determine whether $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ is Hermitian.

FORMULAS YOU NEED

- **Conjugate transpose** A^* : transpose the matrix, then change the sign of every i . (Written A^* , A^H or $A^\dagger$.)
- **Hermitian** means $A^* = A$
- **Unitary** means $A^*A = I$
- Remember $i^2 = -1$, so $-i^2 = +1$

WHY THIS WORKS

A Hermitian matrix is the complex version of a symmetric matrix. For a real matrix, symmetric means $a_{ij} = a_{ji}$. For a complex matrix the right generalisation is $a_{ij} = \text{conjugate of } a_{ji}$ — and that forces the diagonal to be real, because a number equal to its own conjugate has no imaginary part. A unitary matrix is the complex version of an orthogonal matrix: its columns are unit vectors that are perpendicular to each other, which is exactly what $A^*A = I$ says.

SOLUTION — EVERY STEP

1 Part (a) — write down the transpose.

$$A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Swap rows and columns: the entry at position (1,2) moves to (2,1) and vice-versa.

2 Conjugate it — flip the sign of every i .

$$A^* = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

Notice this is the same as the original A . So this matrix happens to be **Hermitian as well** — worth one line in your answer, it costs nothing.

3 Multiply A^*A , entry by entry.

$$A^*A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

Row 1 $\times$ column 1: $(0)(0) + (i)(-i) = -i^2 = 1$

Row 1 $\times$ column 2: $(0)(i) + (i)(0) = 0$

Row 2 $\times$ column 1: $(-i)(0) + (0)(-i) = 0$

Row 2 $\times$ column 2: $(-i)(i) + (0)(0) = -i^2 = 1$

4 State the conclusion.

$$A^*A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Therefore A is unitary. (And Hermitian.)

5 Part (b) — compare each entry with the conjugate of its mirror image.

You do not need to write out the whole conjugate transpose. Just check the three off-diagonal pairs and the diagonal:

- $a_{12} = 7-4i$ and $\text{conj}(a_{21}) = \text{conj}(7+4i) = 7-4i$ ✓
- $a_{13} = -2+5i$ and $\text{conj}(a_{31}) = \text{conj}(-2-5i) = -2+5i$ ✓
- $a_{23} = 3+i$ and $\text{conj}(a_{32}) = \text{conj}(3-i) = 3+i$ ✓
- Diagonal 3, -2, 4 — all real ✓

6 Conclude.

Every entry satisfies $a_{ij} = \text{conj}(a_{ji})$ and the diagonal is real, so $A^* = A$. **Therefore A is Hermitian.**

CHECK YOUR ANSWER

Both answers are YES. If you get “no” for either, you have almost certainly forgotten to conjugate, or conjugated without transposing. Do the two operations in either order — the result is the same — but you must do **both**.

WATCH OUT

- Transposing alone is not enough. $A^T \neq A^*$ when there are complex entries.
- A Hermitian matrix must have a **real diagonal**. If you spot a diagonal entry with an i in it, the answer is no and you can stop there.
- Show the multiplication for the unitary part. Writing only “ $A^*A = I$, so unitary” will not get full marks.

2. Prove T is linear, and find images

Unit III · Q5b · 7 marks · 12 minutes

THE QUESTION

- (a) Prove that $T: R^2 \rightarrow R^3$ given by $T(x, y) = (3x+y, 2y, x-y)$ is linear. Find the images of $(1, 2)$ and $(2, -5)$.
- (b) Show that $T: P_2 \rightarrow P_2$ given by $T(ax^2+bx+c) = (a+b)x^2 + c$ is linear. Find the image of $5x^2+6x+1$.

FORMULAS YOU NEED

A map T is linear if and only if BOTH of these hold for all vectors and all scalars:

- **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
- **Homogeneity:** $T(k\mathbf{u}) = k T(\mathbf{u})$

WHY THIS WORKS

“Linear” means the map does not care whether you combine first and transform, or transform first and combine. The two conditions say exactly that for the two ways of combining vectors: adding them, and scaling them. The proof is always the same shape — take two **general** vectors with subscripts, work out the left side, work out the right side, and show the two expressions are identical. **Do not use numbers**; a check with specific vectors proves nothing.

SOLUTION — EVERY STEP

1 Set up general vectors.

Let $\mathbf{u} = (x_1, y_1)$ and $\mathbf{v} = (x_2, y_2)$, and let k be any scalar. Everything follows from these.

2 Additivity — work out the left-hand side.

First add the vectors: $\mathbf{u} + \mathbf{v} = (x_1+x_2, y_1+y_2)$

Now apply T to that, using the given rule:

$$T(\mathbf{u}+\mathbf{v}) = (3(x_1+x_2) + (y_1+y_2), 2(y_1+y_2), (x_1+x_2) - (y_1+y_2))$$

3 Work out the right-hand side.

$$T(\mathbf{u}) + T(\mathbf{v}) = (3x_1+y_1, 2y_1, x_1-y_1) + (3x_2+y_2, 2y_2, x_2-y_2)$$

Add componentwise:

$$= ((3x_1+y_1) + (3x_2+y_2), 2y_1+2y_2, (x_1-y_1) + (x_2-y_2))$$

Expand the brackets in step 2 and you get exactly this. **So additivity holds.**

4 Homogeneity.

Scale first: $k\mathbf{u} = (kx_1, ky_1)$. Then transform:

$$T(k\mathbf{u}) = (3kx_1 + ky_1, 2ky_1, kx_1 - ky_1)$$

Take the common factor k out of every component:

$$= k(3x_1 + y_1, 2y_1, x_1 - y_1) = kT(\mathbf{u})$$

So homogeneity holds. Both conditions hold, therefore T is linear.

5 Find the images — just substitute.

$$T(1, 2) = (3(1) + 2, 2(2), 1 - 2) = \mathbf{(5, 4, -1)}$$

$$T(2, -5) = (3(2) + (-5), 2(-5), 2 - (-5)) = \mathbf{(1, -10, 7)}$$

Be careful with the double negative in the last component: $2 - (-5) = 2 + 5 = 7$.

6 Part (b) — the polynomial version is the same proof.

Let $p = a_1x^2 + b_1x + c_1$ and $q = a_2x^2 + b_2x + c_2$.

$p + q$ has coefficients $a_1 + a_2$, $b_1 + b_2$, $c_1 + c_2$, so

$$T(p+q) = [(a_1 + a_2) + (b_1 + b_2)]x^2 + (c_1 + c_2)$$

Regroup as $[(a_1 + b_1) + (a_2 + b_2)]x^2 + c_1 + c_2 = T(p) + T(q)$. ✓

Similarly $T(kp) = (ka_1 + kb_1)x^2 + kc_1 = kT(p)$. ✓

7 The image.

For $5x^2 + 6x + 1$ read off $a = 5$, $b = 6$, $c = 1$:

$$T(5x^2 + 6x + 1) = (5 + 6)x^2 + 1 = \mathbf{11x^2 + 1}$$

CHECK YOUR ANSWER

$$\mathbf{(5, 4, -1) \cdot (1, -10, 7) \cdot 11x^2 + 1}$$

A quick sanity check that works for any linear map: $T(\mathbf{0})$ must be $\mathbf{0}$. Here $T(0,0) = (0,0,0)$ ✓. If a map sends $\mathbf{0}$ to something non-zero, it is **not** linear and you can say so immediately.

WATCH OUT

- **Do not verify with numbers.** Showing $T(1,2) + T(3,4) = T(4,6)$ proves nothing. You must use general subscripted vectors.
- Prove **both** conditions. Each is worth marks; showing only additivity typically loses two or three.
- Write the final line explicitly: "Since both conditions are satisfied, T is linear."

3. Least-squares solution of $A\mathbf{x} = \mathbf{b}$

Unit IV · Q8b · 10 marks · 20 minutes

THE QUESTION

Find the least-squares solution of $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

FORMULAS YOU NEED

The normal equations: $A^T A \mathbf{x} = A^T \mathbf{b}$

That is the entire method. Build the two pieces, then solve a small square system.

WHY THIS WORKS

$A\mathbf{x} = \mathbf{b}$ has four equations but only three unknowns, so in general there is **no exact solution** — $\mathbf{b}$ does not lie in the column space of A . The best you can do is find the $\hat{\mathbf{x}}$ that makes $A\hat{\mathbf{x}}$ as close to $\mathbf{b}$ as possible. Geometrically, $A\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto the column space, so the error $\mathbf{b} - A\hat{\mathbf{x}}$ must be perpendicular to every column of A . Writing “perpendicular to every column” in symbols gives $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$, which rearranges to the normal equations. **That one sentence is worth writing in your answer.**

SOLUTION — EVERY STEP

1 Write down A^T .

$$A^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 3 \\ 5 & 0 & 2 & 3 \end{bmatrix}$$

The rows of A^T are the columns of A .

2 Build $A^T A$. It is 3×3 and symmetric, so you only need six numbers.

Each entry is a dot product of two **columns** of A . Call the columns $\mathbf{c}_1 = (1, 1, 1, 1)$, $\mathbf{c}_2 = (3, 1, 1, 3)$, $\mathbf{c}_3 = (5, 0, 2, 3)$.

$$\mathbf{c}_1 \cdot \mathbf{c}_1 = 1 + 1 + 1 + 1 = \mathbf{4}$$

$$\mathbf{c}_1 \cdot \mathbf{c}_2 = 3 + 1 + 1 + 3 = \mathbf{8}$$

$$\mathbf{c}_1 \cdot \mathbf{c}_3 = 5 + 0 + 2 + 3 = \mathbf{10}$$

$$\mathbf{c}_2 \cdot \mathbf{c}_2 = 9 + 1 + 1 + 9 = \mathbf{20}$$

$$\mathbf{c}_2 \cdot \mathbf{c}_3 = 15 + 0 + 2 + 9 = \mathbf{26}$$

$$\mathbf{c}_3 \cdot \mathbf{c}_3 = 25 + 0 + 4 + 9 = \mathbf{38}$$

$$A^T A = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$$

The symmetry is a free check — if your matrix is not symmetric, you have made an arithmetic slip.

3 Build $A^T \mathbf{b}$. Three dot products of the columns with $\mathbf{b}$.

$$\mathbf{c}_1 \cdot \mathbf{b} = 3 + 5 + 7 - 3 = \mathbf{12}$$

$$\mathbf{c}_2 \cdot \mathbf{b} = 3(3) + 1(5) + 1(7) + 3(-3) = 9 + 5 + 7 - 9 = \mathbf{12}$$

$$\mathbf{c}_3 \cdot \mathbf{b} = 5(3) + 0(5) + 2(7) + 3(-3) = 15 + 0 + 14 - 9 = \mathbf{20}$$

$$A^T \mathbf{b} = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$$

4 Write out the system.

$$4x + 8y + 10z = 12 \quad \dots(1)$$

$$8x + 20y + 26z = 12 \quad \dots(2)$$

$$10x + 26y + 38z = 20 \quad \dots(3)$$

5 Eliminate x .

$$(2) - 2 \times (1): (20 - 16)y + (26 - 20)z = 12 - 24$$

$$4y + 6z = -12 \rightarrow 2y + 3z = -6 \quad \dots(4)$$

$$(3) - 2.5 \times (1): (26 - 20)y + (38 - 25)z = 20 - 30$$

$$6y + 13z = -10 \quad \dots(5)$$

6 Eliminate y and solve.

$$(5) - 3 \times (4): 13z - 9z = -10 - (-18)$$

$$4z = 8 \rightarrow z = \mathbf{2}$$

$$\text{Put } z = 2 \text{ into (4): } 2y + 6 = -6 \rightarrow 2y = -12 \rightarrow y = \mathbf{-6}$$

$$\text{Put both into (1): } 4x - 48 + 20 = 12 \rightarrow 4x = 40 \rightarrow x = \mathbf{10}$$

7 State the answer.

$$\hat{\mathbf{x}} = \begin{bmatrix} 10 \\ -6 \\ 2 \end{bmatrix}$$

CHECK YOUR ANSWER

$\hat{\mathbf{x}} = (10, -6, 2)$. Here is a check that takes thirty seconds and confirms all ten marks.

Compute $A\hat{\mathbf{x}} = (2, 4, 8, -2)$. The error is

$$\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}} = (3, 5, 7, -3) - (2, 4, 8, -2) = (1, 1, -1, -1)$$

Now dot $\mathbf{r}$ with each column of A :

- $\mathbf{r} \cdot \mathbf{c}_1 = 1+1-1-1 = 0 \quad \checkmark$
- $\mathbf{r} \cdot \mathbf{c}_2 = 3+1-1-3 = 0 \quad \checkmark$
- $\mathbf{r} \cdot \mathbf{c}_3 = 5+0-2-3 = 0 \quad \checkmark$

All three are zero, so the error really is perpendicular to the column space. **That is the definition of the least-squares solution, so the answer is right.** Writing this check earns marks on its own.

WATCH OUT

- Do **not** try to solve $A\mathbf{x} = \mathbf{b}$ directly. It is inconsistent — four equations, three unknowns. If you row-reduce the augmented matrix you will get a contradiction and panic.
- $A^T A$ must come out symmetric. Use that as a checkpoint before going further.
- Keep the signs on $\mathbf{b}$. The last entry is -3 , and it is easy to drop the minus in the dot products.

4. Prove the t^n -multiplication rule

Unit I · Q1c or Q2b · 7 marks · 15 minutes

THE QUESTION

If $L\{f(t)\} = F(s)$, prove that

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$$

where n is a positive integer.

This has been set word-for-word in all four past papers.

FORMULAS YOU NEED

- **Definition:** $F(s) = \int_0^\infty e^{-st} f(t) dt$
- **Differentiation under the integral sign** (Leibniz's rule): you may move $\frac{d}{ds}$ inside the integral, where it becomes a partial derivative $\frac{\partial}{\partial s}$.
- $\frac{\partial}{\partial s} (e^{-st}) = -t e^{-st}$ — note t is a constant as far as s is concerned.

WHY THIS WORKS

The whole proof rests on one observation: inside the Laplace integral, differentiating with respect to s pulls down a factor of $-t$. So each differentiation multiplies the function by $-t$. Do it n times and you have multiplied by $(-t)^n = (-1)^n t^n$, which is exactly the statement, rearranged. Everything else is bookkeeping. **Write it as a proper induction** — that is what the seven marks are for.

SOLUTION — EVERY STEP

1 Start from the definition.

$$F(s) = \int_0^\infty e^{-st} f(t) dt$$

Every proof in this unit starts on this line. Write it down first, always.

2 Differentiate both sides with respect to s and move the derivative inside.

$$\frac{dF}{ds} = \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt = \int_0^\infty \frac{\partial}{\partial s} [e^{-st}] f(t) dt$$

This step is Leibniz's rule. **Say the name** — examiners look for it.

3 Do the inner differentiation.

$$\frac{\partial}{\partial s}(e^{-st}) = -t e^{-st}$$

so

$$\frac{dF}{ds} = \int_0^\infty (-t)e^{-st}f(t) dt = -\int_0^\infty e^{-st}[t f(t)] dt$$

4 Recognise the integral on the right — it *is* a Laplace transform.

$$\frac{dF}{ds} = -L\{t f(t)\} \rightarrow L\{t f(t)\} = -\frac{dF}{ds} = (-1)^1 \frac{d^1 F}{ds^1}$$

This is the result for $n = 1$ — the base case of the induction.

5 Assume the result holds for some n .

Inductive hypothesis: suppose that for a particular positive integer n ,

$$L\{t^n f(t)\} = \int_0^\infty e^{-st} t^n f(t) dt = (-1)^n \frac{d^n F}{ds^n}$$

6 Differentiate the hypothesis once more.

Left side — same trick as before, another factor of $(-t)$ comes down:

$$\frac{d}{ds} \int_0^\infty e^{-st} t^n f(t) dt = -\int_0^\infty e^{-st} t^{n+1} f(t) dt = -L\{t^{n+1} f(t)\}$$

Right side:

$$\frac{d}{ds} \left[(-1)^n \frac{d^n F}{ds^n} \right] = (-1)^n \frac{d^{n+1} F}{ds^{n+1}}$$

7 Equate and tidy.

$$-L\{t^{n+1} f(t)\} = (-1)^n \frac{d^{n+1} F}{ds^{n+1}}$$

Multiply both sides by -1 :

$$L\{t^{n+1} f(t)\} = (-1)^{n+1} \frac{d^{n+1} F}{ds^{n+1}}$$

This is exactly the statement with n replaced by $n+1$.

8 Conclude.

The result holds for $n = 1$, and whenever it holds for n it also holds for $n+1$. **By the principle of mathematical induction it holds for every positive integer n . ■**

CHECK YOUR ANSWER

Test the formula on something you know. Take $f(t) = 1$, so $F(s) = 1/s$.

The rule with $n = 2$ predicts $L\{t^2\} = (-1)^2 \frac{d^2}{ds^2} (1/s) = \frac{2}{s^3}$, and the standard table also gives $L\{t^2\} = 2!/s^3 = 2/s^3$. ✓

WATCH OUT

- **Do not just do the $n = 1$ case and write “similarly”.** The marks are for the induction. State the hypothesis, do the step, state the conclusion.
- Name Leibniz’s rule when you move the derivative inside the integral.
- Keep track of the sign. It is $(-1)^n$, not (-1) . Each differentiation contributes one factor of -1 .

5. Piecewise function via Heaviside, then its Laplace transform

Unit II · Q3c · 7 marks · 20 minutes

THE QUESTION

Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ in terms of the Heaviside function, and hence find its Laplace transform.

FORMULAS YOU NEED

- **The unit step:** $u(t-a) = 0$ for $t < a$, and 1 for $t \geq a$. It is a switch that turns on at $t = a$.
- **Second shifting theorem:** $L\{g(t-a)u(t-a)\} = e^{-as}G(s)$ where $G(s) = L\{g(t)\}$
- $L\{1\} = 1/s$, $L\{t\} = 1/s^2$, $L\{t^2\} = 2/s^3$

WHY THIS WORKS

Think of the function as switching. It starts as t^2 . At $t = 2$ it must **stop** being t^2 and **start** being $4t$. You achieve that by adding the difference $(4t - t^2)$ switched on at $t = 2$. At $t = 4$ you add the next difference, $(8 - 4t)$, switched on at $t = 4$. Each new piece cancels the old one and installs the new one.

The one trap: the second shifting theorem only applies when the thing multiplying $u(t-a)$ is written as a function of $(t-a)$, not of t . So there is always a rewriting step, and that is where most marks are lost.

SOLUTION — EVERY STEP

1 Write the function as a sum of switches.

General pattern: $f = (\text{first piece}) + (\text{second} - \text{first})u(t-a_1) + (\text{third} - \text{second})u(t-a_2) + \dots$

$$f(t) = t^2 + (4t - t^2)u(t-2) + (8 - 4t)u(t-4)$$

Sanity check. For t between 0 and 2 both switches are off, so $f = t^2$ ✓. For t between 2 and 4 the first switch is on: $f = t^2 + 4t - t^2 = 4t$ ✓. For $t > 4$ both are on: $f = 4t + 8 - 4t = 8$ ✓.

2 Transform the first term.

$$L\{t^2\} = 2/s^3$$

No switch on this one, so no shifting is needed.

3 Rewrite the second bracket in terms of $(t - 2)$.

Substitute $t = \tau + 2$ (so $\tau = t - 2$):

$$\begin{aligned} 4t - t^2 &= 4(\tau+2) - (\tau+2)^2 \\ &= 4\tau + 8 - (\tau^2 + 4\tau + 4) = 4\tau + 8 - \tau^2 - 4\tau - 4 = 4 - \tau^2 \end{aligned}$$

So the term is $g(t-2)u(t-2)$ with $g(\tau) = 4 - \tau^2$.

4 Transform it.

$$G(s) = L\{4 - \tau^2\} = \frac{4}{s} - \frac{2}{s^3}$$

Apply the second shifting theorem — multiply by e^{-2s} :

$$e^{-2s}\left(\frac{4}{s} - \frac{2}{s^3}\right)$$

5 Rewrite the third bracket in terms of $(t - 4)$.

Substitute $t = \tau + 4$:

$$8 - 4t = 8 - 4(\tau+4) = 8 - 4\tau - 16 = -8 - 4\tau$$

So $g(\tau) = -8 - 4\tau$.

6 Transform it.

$$G(s) = -\frac{8}{s} - \frac{4}{s^2}$$

Multiply by e^{-4s} :

$$e^{-4s}\left(-\frac{8}{s} - \frac{4}{s^2}\right)$$

7 Add the three pieces.

$$L\{f(t)\} = \frac{2}{s^3} + e^{-2s}\left(\frac{4}{s} - \frac{2}{s^3}\right) + e^{-4s}\left(-\frac{8}{s} - \frac{4}{s^2}\right)$$

CHECK YOUR ANSWER

Two quick checks.

1. The switch check in step 1. Put a value of t in each interval into your Heaviside expression and confirm you get back the original piece. If that works, the hard half is right.

2. Every exponential must match its switch. The term switched on at $t = 2$ carries e^{-2s} ; the one at $t = 4$ carries e^{-4s} . If an exponent does not match its switch point, you have mixed up a term.

WATCH OUT

- **The rewriting step is compulsory.** Writing $L\{(4t-t^2)u(t-2)\} = e^{-2s}L\{4t-t^2\}$ is **wrong** and will cost you most of the marks. You must convert to a function of $\tau = t-2$ first.
- Use **differences**, not the pieces themselves: it is $(4t - t^2)$, not $4t$.
- Expand $(\tau+2)^2$ carefully — it is $\tau^2 + 4\tau + 4$, and the 4τ is what cancels.

6. Given $L\{\sin \sqrt{t}\}$, show the $\cos \sqrt{t}$ result

Unit I · Q1b · 4 marks · 12 minutes

THE QUESTION

Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, show that

$$L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$$

FORMULAS YOU NEED

- **Transform of a derivative:** $L\{f'(t)\} = s F(s) - f(0)$
- **Chain rule:** $\frac{d}{dt}(\sin \sqrt{t}) = \cos \sqrt{t} \times \frac{d}{dt}(\sqrt{t})$
- $\frac{d}{dt}(\sqrt{t}) = \frac{1}{2\sqrt{t}}$

WHY THIS WORKS

You are given a transform and asked for a different one. The trick is to spot that the **function you want is just a multiple of the derivative of the function you were given**. Differentiating $\sin \sqrt{t}$ produces $\cos \sqrt{t}$ over $2\sqrt{t}$ — almost exactly the target, off by a factor of 2. So instead of integrating anything, you use the derivative rule, which turns the problem into one multiplication by s .

SOLUTION — EVERY STEP

- 1 Name the function and note its value at zero.**

Let $f(t) = \sin \sqrt{t}$. Then $F(s) = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$ is what you were given.

$f(0) = \sin 0 = 0$. Note this now — it is what makes the next step clean.

- 2 Differentiate f using the chain rule.**

$$f'(t) = \cos \sqrt{t} \times \frac{1}{2\sqrt{t}} = \frac{\cos \sqrt{t}}{2\sqrt{t}}$$

- 3 Spot the relationship to the target.**

$$\frac{\cos \sqrt{t}}{\sqrt{t}} = 2 \times \frac{\cos \sqrt{t}}{2\sqrt{t}} = 2 f'(t)$$

The function you want is exactly twice the derivative. That is the whole idea of the question.

4 Take the Laplace transform of both sides.

$$L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = 2 L\{f(t)\}$$

Apply the derivative rule $L\{f\} = sF(s) - f(0)$:

$$= 2[s F(s) - f(0)] = 2s F(s) - 0 = 2s F(s)$$

5 Substitute the given transform and simplify.

$$= 2s \times \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$$

The 2 cancels the 2, and $s \div s^{3/2} = s^{1-3/2} = s^{-1/2}$:

$$= \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s} \quad \blacksquare$$

CHECK YOUR ANSWER

The exponential factor $e^{-1/4s}$ is untouched throughout — it should appear unchanged in your final line. Only the power of s and the constant change. If your exponential has moved, you have differentiated the wrong thing.

The index arithmetic: $s^1 \div s^{3/2} = s^{-1/2} = 1/\sqrt{s} = 1/s^{1/2}$. ✓

WATCH OUT

- **Do not try to derive $L\{\sin \sqrt{t}\}$ from scratch.** It is *given*. Deriving it needs a series expansion and will eat your whole exam.
- You must state $f(0) = 0$. If you leave the $-f(0)$ term dangling you lose a mark, even though it is zero.
- The factor is **2**, not $\frac{1}{2}$. Differentiating gives a 2 in the *denominator*, so you multiply by 2 to remove it.

7. The 2-mark definitions

Units I and II · Q1a, Q2a, Q3a, Q4a · 2 marks each · 10 minutes

THE QUESTION

One of these opens **every** question in Units I and II, so you get one whichever you choose:

- Define the Laplace transform of a function.
- Write the Laplace transform of a periodic function.
- Define a periodic function with an example.
- Define the unit step function and represent it graphically.
- Define the Dirac-delta function and sketch its graph.

WHAT TO WRITE — LEARN THESE FIVE ANSWERS WORD FOR WORD

1. Laplace transform. For a function $f(t)$ defined for $t \geq 0$,

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$$

provided the integral converges.

2. Periodic function. f is periodic with period T if $f(t+T) = f(t)$ for all t . Example: $\sin t$, with period 2π .

3. Laplace transform of a periodic function of period T :

$$L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

4. Unit step (Heaviside) function.

$$u(t-a) = 0 \text{ for } t < a, \quad u(t-a) = 1 \text{ for } t \geq a$$

Graph: flat along zero, then a vertical jump of height 1 at $t = a$, then flat at 1. Also useful:

$$L\{u(t-a)\} = e^{-as}/s.$$

5. Dirac-delta (unit impulse) function. $\delta(t-a)$ is the limit of a rectangular pulse of width ε and height $1/\varepsilon$ centred at $t = a$, as $\varepsilon \rightarrow 0$. It satisfies

$$\delta(t-a) = 0 \text{ for } t \neq a, \quad \text{and} \quad \int_{-\infty}^\infty \delta(t-a) dt = 1$$

Graph: a single vertical arrow at $t = a$, labelled with area 1. Also useful: $L\{\delta(t-a)\} = e^{-as}$.

WATCH OUT

- These are worth **4 marks per paper** (one in Unit I, one in Unit II) for about five sentences. Do not skip them because they look trivial.
- When the question says “represent graphically” or “sketch”, **draw the graph**. Half the mark is the picture. Label the axes and the point $t = a$.
- For the Laplace transform definition, add “provided the integral converges”. It shows you know it does not always exist.

8. Solving ODEs and systems of ODEs by Laplace transforms

Unit II · Q3d or Q4d · 7 marks · 22 minutes

THE QUESTION

(a) **A system.** $\frac{dx}{dt} - 2y = \cos 2t$, $\frac{dy}{dt} + 2x = \sin 2t$, given $x(0) = 1$, $y(0) = 0$.

(b) **A single ODE.** $y'' + 4y' + 3y = e^{-t}$, given $y(0) = 0$, $y'(0) = 0$.

(c) **A circuit.** A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show the current is $\frac{E}{R - aL} (e^{-at} - e^{-Rt/L})$.

FORMULAS YOU NEED

- $L\{y'\} = sY - y(0)$
- $L\{y''\} = s^2Y - sy(0) - y'(0)$
- $L\{\cos at\} = s/(s^2 + a^2)$, $L\{\sin at\} = a/(s^2 + a^2)$
- $L\{e^{-at}\} = 1/(s + a)$, $L\{te^{-at}\} = 1/(s + a)^2$

WHY THIS WORKS

A differential equation is hard because it mixes a function with its derivatives. The Laplace transform converts derivatives into **multiplication by s** , which turns the whole differential equation into an ordinary algebraic one. You solve the algebra for $Y(s)$, then transform back. The initial conditions are not an afterthought — they enter automatically at the transform step, which is why this method is often faster than the classical one.

For a system, both equations transform, and you end up with two linear equations in $X(s)$ and $Y(s)$. Solve them the same way you would solve any pair of simultaneous equations, then invert each.

SOLUTION — EVERY STEP

1 (a) Transform both equations.

Write $X = L\{x\}$ and $Y = L\{y\}$. Using $x(0) = 1$ and $y(0) = 0$:

$$sX - 1 - 2Y = \frac{s}{s^2 + 4} \rightarrow sX - 2Y = 1 + \frac{s}{s^2 + 4} \quad \dots(i)$$

$$sY + 2X = \frac{2}{s^2 + 4} \rightarrow 2X + sY = \frac{2}{s^2 + 4} \quad \dots(ii)$$

2 Eliminate Y.

Multiply (i) by s and (ii) by 2, then add — the Y terms cancel:

$$s \times (i): \quad s^2 X - 2sY = s + \frac{s^2}{s^2+4}$$

$$2 \times (ii): \quad 4X + 2sY = \frac{4}{s^2+4}$$

$$\text{Add: } (s^2+4)X = s + \frac{s^2+4}{s^2+4} = s+1$$

The right-hand side simplified beautifully. So

$$X = \frac{s+1}{s^2+4}$$

3 Invert to get $x(t)$.

Split the fraction:

$$X = \frac{s}{s^2+4} + \frac{1}{s^2+4} = \frac{s}{s^2+4} + \frac{1}{2} \cdot \frac{2}{s^2+4}$$

The second term needed a 2 on top to match the sin formula, so we put in $\frac{1}{2}$ outside to compensate.

$$x(t) = \cos 2t + \frac{1}{2} \sin 2t$$

4 Back-substitute for Y.

$$\text{From (ii): } sY = \frac{2}{s^2+4} - 2X$$

$$= \frac{2}{s^2+4} - \frac{2(s+1)}{s^2+4} = \frac{2-2s-2}{s^2+4} = \frac{-2s}{s^2+4}$$

Divide by s :

$$Y = \frac{-2}{s^2+4} \rightarrow y(t) = -\sin 2t$$

5 (b) Transform the single ODE.

With $y(0) = 0$ and $y'(0) = 0$, both boundary terms vanish:

$$s^2 Y + 4sY + 3Y = \frac{1}{s+1}$$

$$(s^2 + 4s + 3)Y = \frac{1}{s+1} \rightarrow (s+1)(s+3)Y = \frac{1}{s+1}$$

$$Y = \frac{1}{(s+1)^2(s+3)}$$

Note the repeated factor — the $(s+1)$ from the forcing term meets the $(s+1)$ from the characteristic polynomial and gives a square.

6 Partial fractions.

$$\frac{1}{(s+1)^2(s+3)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+3}$$

Multiply through by $(s+1)^2(s+3)$:

$$1 = A(s+1)(s+3) + B(s+3) + C(s+1)^2$$

Put $s = -1$: $1 = B(2) \rightarrow \mathbf{B = 1/2}$

Put $s = -3$: $1 = C(4) \rightarrow \mathbf{C = 1/4}$

Put $s = 0$: $1 = 3A + 3(1/2) + 1/4 \rightarrow 3A = 1 - 7/4 = -3/4 \rightarrow \mathbf{A = -1/4}$

7 Invert term by term.

$$Y = -\frac{1}{4} \cdot \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{(s+1)^2} + \frac{1}{4} \cdot \frac{1}{s+3}$$

$$y(t) = -\frac{1}{4} e^{-t} + \frac{1}{2} t e^{-t} + \frac{1}{4} e^{-3t}$$

8 (c) The circuit — write the equation first.

Kirchhoff's voltage law for a series LR circuit gives

$$L \frac{di}{dt} + Ri = Ee^{-at}, \quad i(0) = 0$$

Transform, using $i(0) = 0$:

$$L(sI) + RI = \frac{E}{s+a} \rightarrow I = \frac{E}{(s+a)(Ls+R)}$$

9 Split and invert.

Take L out of the second bracket: $I = \frac{E/L}{(s+a)(s+R/L)}$

Use the standard split $\frac{1}{(s+p)(s+q)} = \frac{1}{q-p} \left[\frac{1}{s+p} - \frac{1}{s+q} \right]$ with $p = a$, $q = R/L$. Since $q - p = (R - aL)/L$, the L 's cancel and the constant becomes $E/(R - aL)$:

$$I = \frac{E}{R - aL} \left[\frac{1}{s+a} - \frac{1}{s+R/L} \right]$$

$$i(t) = \frac{E}{R - aL} (e^{-at} - e^{-Rt/L}) \quad \blacksquare$$

CHECK YOUR ANSWER

Always substitute back. It is fast and it catches sign errors.

(a) $x = \cos 2t + \frac{1}{2}\sin 2t$, $y = -\sin 2t$.

$$x' = -2\sin 2t + \cos 2t, \text{ so } x' - 2y = -2\sin 2t + \cos 2t + 2\sin 2t = \cos 2t \quad \checkmark$$

$$y' = -2\cos 2t, \text{ so } y' + 2x = -2\cos 2t + 2\cos 2t + \sin 2t = \sin 2t \quad \checkmark$$

$$x(0) = 1 \quad \checkmark, \quad y(0) = 0 \quad \checkmark$$

(b) $y(0) = -\frac{1}{4} + 0 + \frac{1}{4} = 0 \quad \checkmark$. And $y'(0) = \frac{1}{4} + \frac{1}{2} - \frac{3}{4} = 0 \quad \checkmark$. Both initial conditions recovered.

WATCH OUT

- **Put the initial conditions in at the transform step**, not at the end. That is the whole advantage of the method; forgetting them gives you a general solution with unknown constants and no marks.
- $L\{y''\}$ has **two** boundary terms: $-sy(0)$ **and** $-y'(0)$. Dropping the second is the most common error in this question.
- When the forcing term shares a root with the characteristic polynomial (as in (b)), you get a **repeated factor** and the partial fractions need both $A/(s+1)$ and $B/(s+1)^2$.
- In a system, solve for one transform completely, then **substitute back** into the simpler of the two equations to get the other. Do not eliminate twice.

9. Singular Value Decomposition

Unit V · Q9c · 10 marks · 22 minutes

THE QUESTION

Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$

FORMULAS YOU NEED

Goal: write $A = U\Sigma V^T$.

- Form $A^T A$ (always square and symmetric).
- Its eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ are all ≥ 0 . The **singular values** are $\sigma_i = \sqrt{\lambda_i}$.
- The unit eigenvectors of $A^T A$ are the columns of V (largest eigenvalue first).
- The columns of U come from $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\sigma_i}$
- Σ is the same shape as A , with $\sigma_1, \sigma_2, \dots$ down the diagonal and zeros elsewhere.

WHY THIS WORKS

Any matrix, even a non-square one, can be broken into three simple actions: a rotation (V^T), a stretch along the axes (Σ), and another rotation (U). Finding those pieces directly is hard, but $A^T A$ is symmetric, and symmetric matrices are easy — real eigenvalues, orthogonal eigenvectors. So you find V from $A^T A$, take square roots for the stretch factors, and then U falls out of $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$.

There is no calculus in this question at all — it is eigenvectors, done twice.

SOLUTION — EVERY STEP

1 Compute $A^T A$.

$$A^T = \begin{bmatrix} 1 & 3 \\ 1 & -3 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 1 & 3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$$

Entry (1,1): $(1)(1) + (3)(3) = 10$

Entry (1,2): $(1)(1) + (3)(-3) = 1 - 9 = -8$

Entry (2,1): $(1)(1) + (-3)(3) = -8$

Entry (2,2): $(1)(1) + (-3)(-3) = 10$

$$A^T A = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$$

It came out symmetric, which is the check that you multiplied correctly.

2 Find its eigenvalues.

$$\det\begin{pmatrix} 10-\lambda & -8 \\ -8 & 10-\lambda \end{pmatrix} = (10-\lambda)^2 - 64 = 0$$

So $(10 - \lambda)^2 = 64$, giving $10 - \lambda = \pm 8$:

$$\lambda_1 = 18, \quad \lambda_2 = 2$$

Shortcut worth knowing: for a 2×2 of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ the eigenvalues are always $a + b$ and $a - b$.

3 Take square roots to get the singular values.

$$\sigma_1 = \sqrt{18} = 3\sqrt{2}, \quad \sigma_2 = \sqrt{2}$$

Always list them **largest first**.

4 Find the eigenvectors of $A^T A$ — these give V .

For $\lambda = 18$: $(A^T A - 18I)\mathbf{v} = \mathbf{0}$ gives

$$-8v_1 - 8v_2 = 0 \rightarrow v_1 = -v_2 \rightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\textbf{For } \lambda = 2: 8v_1 - 8v_2 = 0 \rightarrow v_1 = v_2 \rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Both have length $\sqrt{2}$, so normalise by dividing by $\sqrt{2}$:

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

5 Get the columns of U from $\mathbf{u}_i = A\mathbf{v}_i / \sigma_i$.

$$\textbf{First column. } A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 3+3 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$\text{So } A\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 6 \end{bmatrix}, \text{ and dividing by } \sigma_1 = 3\sqrt{2}:$$

$$\mathbf{u}_1 = \frac{1}{\sqrt{2} \cdot 3\sqrt{2}} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\textbf{Second column. } A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1 \\ 3-3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2} \cdot \sqrt{2}} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

6 Assemble the three matrices.

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$A = U\Sigma V^T$$

CHECK YOUR ANSWER

Multiply the three back together. Take $V^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

$$\text{First } \Sigma V^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}$$

$$\text{Then } U(\Sigma V^T) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix} = A \quad \checkmark$$

Two faster checks if you are short of time: the singular values squared must sum to the sum of all a_{ij}^2 — here $18 + 2 = 20$ and $1+1+9+9 = 20 \checkmark$. And $\sigma_1\sigma_2 = |\det A|$ — here $3\sqrt{2} \cdot \sqrt{2} = 6$ and $|\det A| = |-3-3| = 6 \checkmark$.

WATCH OUT

- Use $A^T A$, **not** AA^T , to find V . (AA^T would give you U directly, but then the columns can come out with the wrong signs.)
- The singular values are the **square roots** of the eigenvalues. Writing $\Sigma = \text{diag}(18, 2)$ is a very common and very costly slip.
- Order matters: largest singular value first, and the columns of V must be in the matching order.
- Do not forget to **normalise** the eigenvectors before putting them into V .

10. Orthogonal diagonalization

Unit V · Q10c · 10 marks · 25 minutes

THE QUESTION

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

This exact matrix appeared in three of the four past papers.

FORMULAS YOU NEED

- Eigenvalues from $\det(A - \lambda I) = 0$
- Eigenvectors from $(A - \lambda I)\mathbf{x} = \mathbf{0}$
- For a **symmetric** matrix, eigenvectors belonging to **different** eigenvalues are automatically orthogonal.
- Normalise each eigenvector (divide by its length) and use them as the columns of P . Then $P^T A P = D = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$.

WHY THIS WORKS

Ordinary diagonalization gives $P^{-1}AP = D$, and inverting P is work. But when A is symmetric you can choose the eigenvectors to be mutually perpendicular and of unit length. For such a matrix the inverse is just the transpose, so $P^{-1} = P^T$ and no inversion is needed. That is the whole point of the word “orthogonally”.

Here all three eigenvalues are different, so the orthogonality is free — you do not need Gram-Schmidt. **Say this in your answer.** You would only need Gram-Schmidt if an eigenvalue repeated.

SOLUTION — EVERY STEP

1 Form the characteristic equation.

$$\det \begin{bmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix} = 0$$

Expanding along the first row gives

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

Shortcut for checking: the coefficient 11 must equal the trace ($3 + 5 + 3 = 11$ ✓), and the constant 36 must equal $\det A$ (✓).

2 Solve it — try small factors of 36.

Try $\lambda = 2$: $8 - 44 + 72 - 36 = 0$ ✓, so $(\lambda - 2)$ is a factor.

Divide out: $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = (\lambda - 2)(\lambda^2 - 9\lambda + 18) = (\lambda - 2)(\lambda - 3)(\lambda - 6)$

$\lambda = 2, 3, 6$

All different — so the eigenvectors will come out orthogonal on their own.

3 Eigenvector for $\lambda = 2$.

$$A - 2I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Row 3 is the same as row 1, so drop it. Add row 1 to row 2: $(0, 2, 0)$, giving $2y = 0$, so $y = 0$.

Row 1 then says $x - 0 + z = 0$, so $x = -z$. Taking $z = -1$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

4 Eigenvector for $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

Row 3: $x - y = 0$, so $x = y$.

Row 1: $-y + z = 0$, so $z = y$.

Taking $y = 1$:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Check with row 2: $-1 + 2 - 1 = 0$ ✓

5 Eigenvector for $\lambda = 6$.

$$A - 6I = \begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

Row 2 gives $x + y + z = 0$. Row 3 gives $x - y - 3z = 0$.

Subtract the second from the first: $2y + 4z = 0$, so $y = -2z$.

Then $x = -y - z = 2z - z = z$. Taking $z = 1$:

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Check with row 1: $-3(1) - 1(-2) + 1(1) = -3 + 2 + 1 = 0$ ✓

6 Verify orthogonality — this earns marks, so write it out.

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = (1)(1) + (0)(1) + (-1)(1) = \mathbf{0} \quad \checkmark$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = (1)(1) + (0)(-2) + (-1)(1) = \mathbf{0} \quad \checkmark$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = (1)(1) + (1)(-2) + (1)(1) = \mathbf{0} \quad \checkmark$$

All three are mutually perpendicular, as guaranteed because A is symmetric with distinct eigenvalues. No Gram-Schmidt is required.

7 Normalise and assemble P .

$$\text{Lengths: } |\mathbf{v}_1| = \sqrt{1+0+1} = \sqrt{2}, \quad |\mathbf{v}_2| = \sqrt{3}, \quad |\mathbf{v}_3| = \sqrt{1+4+1} = \sqrt{6}$$

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$$

8 State the conclusion.

$$P^T A P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

The eigenvalues appear on the diagonal **in the same order as their eigenvectors sit in P .**

CHECK YOUR ANSWER

Three checks, each ten seconds.

1. **Trace:** the eigenvalues must add up to the trace. $2 + 3 + 6 = 11$, and $3 + 5 + 3 = 11 \quad \checkmark$

2. **Determinant:** the eigenvalues must multiply to $\det A$. $2 \times 3 \times 6 = 36 \quad \checkmark$

3. **Each eigenvector:** compute $A\mathbf{v}$ and confirm it equals $\lambda\mathbf{v}$. For example $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 3$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \checkmark$$

WATCH OUT

- **Normalise.** Putting the raw eigenvectors into P gives $P^{-1}AP = D$ but **not** $P^TAP = D$, and the question said *orthogonally*.
- Keep the order consistent. If $\mathbf{v}_1$ (for $\lambda=2$) is the first column, then 2 must be the first diagonal entry.
- Do not waste time on Gram-Schmidt here. State that the eigenvalues are distinct and the orthogonality is automatic.
- If you get an eigenvalue wrong, everything after it is wrong. **Use the trace check before going on.**

11. Span, basis and linear independence

Unit III · Q5a · 6 marks · 12 minutes

THE QUESTION

- (a) Determine whether $\left\{ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix} \right\}$ is a basis of M_{22} , the space of all 2×2 matrices.
- (b) Do $\{(1,2,3), (-1,-1,0), (2,5,4)\}$ span R^3 ?
- (c) Is $(1,-2)$ a linear combination of $(2,4)$ and $(3,6)$?

FORMULAS YOU NEED

- **Basis** = a set that is linearly independent **and** spans the space. For a space of dimension n , any n independent vectors form a basis automatically.
- **Independence test**: stack the vectors as the rows (or columns) of a square matrix and take the **determinant**. Non-zero $\rightarrow$ independent. Zero $\rightarrow$ dependent.
- $\dim R^3 = 3$, $\dim M_{22} = 4$, $\dim P_2 = 3$

WHY THIS WORKS

The key move is **coordinates**. A 2×2 matrix has four independent entries, so it behaves exactly like a vector in R^4 ; a quadratic behaves like a vector in R^3 . Once you flatten each object into a plain list of numbers, every version of this question becomes the same determinant calculation. **Do not be thrown by the matrices or polynomials** — flatten and carry on.

SOLUTION — EVERY STEP

- 1 (a) Flatten each matrix into a list of four numbers, reading across the rows.

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \rightarrow (1, 2, 0, 1) \quad \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix} \rightarrow (3, 4, 1, 1)$$
$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \rightarrow (1, 2, 1, 1) \quad \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix} \rightarrow (0, 2, 1, 2)$$

You are now working in R^4 . Four vectors, dimension four — so this is purely a question of independence.

2 Stack them as rows and take the determinant.

$$\det \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 4 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 0 & 2 & 1 & 2 \end{bmatrix}$$

Row-reduce to make the first column mostly zeros. $R_2 - 3R_1$ and $R_3 - R_1$ do not change the determinant:

$$= \det \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -2 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 2 \end{bmatrix}$$

3 Expand and finish.

Expanding along the first column leaves the 3×3 determinant

$$\det \begin{bmatrix} -2 & 1 & -2 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Expand along the middle row (it has two zeros):

$$= 1 \times \det \begin{bmatrix} -2 & -2 \\ 2 & 2 \end{bmatrix} = (-2)(2) - (-2)(2) = -4 + 4 = \mathbf{0}$$

4 State the conclusion.

The determinant is zero, so the four matrices are linearly dependent. Therefore they do NOT form a basis of M_{22} .

You could add that the set spans only a 3-dimensional subspace of the 4-dimensional space M_{22} , so it cannot be a basis.

5 (b) The span question — same method.

$$\det \begin{bmatrix} 1 & 2 & 3 \\ -1 & -1 & 0 \\ 2 & 5 & 4 \end{bmatrix}$$

Expanding: $1(-4 - 0) - 2(-4 - 0) + 3(-5 + 2) = -4 + 8 - 9 = \mathbf{-5}$

Non-zero, so the three vectors are independent. Three independent vectors in a 3-dimensional space **do span $\mathbf{R}^3$** . If the question also asks you to express a particular vector, solve the system $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{b}$.

6 (c) The linear-combination question — solve, do not take a determinant.

You want scalars with $c_1(2,4) + c_2(3,6) = (1,-2)$:

$$2c_1 + 3c_2 = 1 \quad 4c_1 + 6c_2 = -2$$

The second equation is exactly twice the left side of the first, so it demands $2(1) = -2$, i.e. $2 = -2$. **Contradiction — so $(1,-2)$ is NOT a linear combination of the two.**

Geometrically: $(2,4)$ and $(3,6)$ are parallel, so they span only a line, and $(1,-2)$ is not on it.

CHECK YOUR ANSWER

(a) determinant 0 → NOT a basis. (b) determinant -5 → they DO span. (c) NOT a linear combination.

Fast sanity check for (a): look for an obvious relation among the flattened vectors. Here $(1,2,0,1) + (1,2,1,1) = (2,4,1,2)$ and $(3,4,1,1) - (0,2,1,2) = (3,2,0,-1)$ — not obvious, which is why the determinant is the reliable route. **Trust the determinant.**

WATCH OUT

- **Count first.** If the number of vectors is not equal to the dimension of the space, they cannot be a basis regardless of independence — too few cannot span, too many must be dependent. Say so and save yourself the work.
- Flatten consistently. Read every matrix across the rows, or every one down the columns, but do not mix.
- A determinant of zero means **dependent**, which means **not a basis**. Do not let the answer “no” make you doubt correct arithmetic — here the answer really is no.

12. The rotation-operator proof

Unit III · Q5c · 7 marks · 12 minutes

THE QUESTION

Show that the linear operator $T: R^2 \rightarrow R^2$ defined by $T(\mathbf{x}) = A\mathbf{x}$ rotates a vector $\mathbf{x}$ through an angle θ about the origin, where

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

FORMULAS YOU NEED

- **Polar form:** any vector in R^2 can be written $\mathbf{x} = (r\cos\varphi, r\sin\varphi)$, where $r = |\mathbf{x}|$ is its length and φ is the angle it makes with the positive x -axis.
- **Compound angle formulae:**
 $\cos(\theta + \varphi) = \cos\theta\cos\varphi - \sin\theta\sin\varphi$
 $\sin(\theta + \varphi) = \sin\theta\cos\varphi + \cos\theta\sin\varphi$

WHY THIS WORKS

A rotation is defined by two properties: it **keeps the length** of a vector the same, and it **increases the angle** by a fixed amount. So the proof has to produce both facts. The trick is to write $\mathbf{x}$ in polar form, because then the angle is visible as φ . Multiplying by A and applying the compound-angle formulae turns φ into $\theta + \varphi$ while leaving r untouched — and that *is* a rotation. **Twelve minutes to learn, seven marks, no calculus.**

SOLUTION — EVERY STEP

1 State that T is linear.

For any matrix A , the map $\mathbf{x} \mapsto A\mathbf{x}$ satisfies $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ and $A(k\mathbf{u}) = k(A\mathbf{u})$ by the rules of matrix multiplication. **So T is a linear operator.** One line, but say it.

2 Write a general vector in polar form.

Let $\mathbf{x}$ have length r and make an angle φ with the positive x -axis:

$$\mathbf{x} = \begin{bmatrix} r\cos\varphi \\ r\sin\varphi \end{bmatrix}$$

This is completely general — every vector in R^2 can be written this way.

3 Multiply by A .

$$A\mathbf{x} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} r\cos\varphi \\ r\sin\varphi \end{bmatrix}$$

Taking the two rows in turn:

$$= \begin{bmatrix} r\cos\theta\cos\varphi - r\sin\theta\sin\varphi \\ r\sin\theta\cos\varphi + r\cos\theta\sin\varphi \end{bmatrix}$$

4 Take out the common factor r .

$$= r \begin{bmatrix} \cos \theta \cos \varphi - \sin \theta \sin \varphi \\ \sin \theta \cos \varphi + \cos \theta \sin \varphi \end{bmatrix}$$

Now look carefully at the two entries — each is exactly one of the compound-angle formulae.

5 Apply the compound-angle formulae.

$$A\mathbf{x} = r \begin{bmatrix} \cos(\theta + \varphi) \\ \sin(\theta + \varphi) \end{bmatrix}$$

6 Read off the two facts that make it a rotation.

Length. $|A\mathbf{x}| = r\sqrt{\cos^2(\theta+\varphi) + \sin^2(\theta+\varphi)} = r \times 1 = r = |\mathbf{x}|$.

The length is unchanged.

Angle. The original vector had angle φ ; the image has angle $\theta + \varphi$. **The angle has increased by exactly θ .**

7 Conclude.

Since T preserves length and increases the angle of every vector by θ , **T rotates $\mathbf{x}$ through an angle θ about the origin.** ■

CHECK YOUR ANSWER

Test it on a case you can see. Take $\theta = 90^\circ$, so $\cos \theta = 0$ and $\sin \theta = 1$:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The vector pointing along the x -axis has become the vector pointing along the y -axis — a 90° anticlockwise turn, exactly as claimed. ✓

WATCH OUT

- **Do not use a specific vector** such as $(1,0)$ as your proof. That demonstrates one case; the question says “a vector $\mathbf{x}$ ”, meaning any vector. Polar form is what makes it general.
- **You must mention the length.** Showing the angle increases by θ is only half a rotation — a spiral does that too. The proof needs $|A\mathbf{x}| = |\mathbf{x}|$.
- Watch the minus sign in the top-right entry of A . If you drop it you get a reflection, not a rotation.

13. Positive-definite check

Unit V · Q10a · 5 marks · 8 minutes · overtime item

THE QUESTION

Check whether $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ is positive definite or not.

FORMULAS YOU NEED

Sylvester's criterion. A symmetric matrix is **positive definite** if and only if **every** leading principal minor is strictly greater than zero.

The leading principal minors are the determinants of the top-left 1×1 , 2×2 , 3×3 ... blocks.

- positive definite $\Leftrightarrow$ all minors > 0
- positive **semi**-definite $\Leftrightarrow$ all eigenvalues ≥ 0 (some may be zero)

WHY THIS WORKS

“Positive definite” means $\mathbf{x}^T A \mathbf{x} > 0$ for every non-zero $\mathbf{x}$ — the matrix always produces a positive number. Checking that directly for all $\mathbf{x}$ is impossible, so Sylvester's criterion gives you a finite test: just three determinants for a 3×3 . **The word “every” matters** — one minor that is zero or negative kills it.

SOLUTION — EVERY STEP

1 First minor: the top-left entry.

$$D_1 = |1| = 1 \quad (> 0 \checkmark)$$

2 Second minor: the top-left 2×2 block.

$$D_2 = \det \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} = (1)(4) - (-2)(-2) = 4 - 4 = 0$$

This is already fatal — the criterion needs strictly greater than zero. But compute the third minor too, because it tells you what the matrix actually is.

3 Third minor: the whole determinant.

Look at the matrix: **row 3 is identical to row 1**. A determinant with two equal rows is zero.

$$D_3 = \det A = 0$$

4 State the conclusion properly.

The minors are 1, 0, 0. Not all are positive, so

A is NOT positive definite.

Now say what it **is**. Notice that

$$A = \mathbf{v}\mathbf{v}^T \text{ where } \mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

so for any $\mathbf{x}$,

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T \mathbf{v} \mathbf{v}^T \mathbf{x} = (\mathbf{v}^T \mathbf{x})^2 \geq 0$$

Therefore A is positive SEMI-definite. Its eigenvalues are 0, 0 and 6.

CHECK YOUR ANSWER

Not positive definite; positive semi-definite.

Confirm with the eigenvalues. Trace = 1 + 4 + 1 = 6 and det = 0, and since the matrix has rank 1 (all rows are multiples of (1, -2, 1)), two eigenvalues are 0 and the third must be 6 to make the trace work. All are ≥ 0 but not all > 0 — that is exactly semi-definite. ✓

WATCH OUT

- **Do not stop at “no”.** Naming it positive semi-definite, and giving a reason, is worth the difference between three marks and five.
- The minors must be the **leading** ones — always taken from the top-left corner. Any other 2×2 block is not part of the test.
- Spot repeated or proportional rows early. Here row 3 = row 1 tells you det = 0 before you calculate anything.

14. The four fundamental subspaces

Unit IV · Q7c · 7 marks · 15 minutes · overtime item

THE QUESTION

Find the dimension and a basis for the four fundamental subspaces of

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

This exact matrix appeared in three of the four past papers.

FORMULAS YOU NEED

For an $m \times n$ matrix of rank r — here $m = 3$, $n = 4$:

- **Column space** $C(A)$ lives in R^m , dimension r . Basis = the **pivot columns of the original A** .
- **Row space** $C(A^T)$ lives in R^n , dimension r . Basis = the non-zero rows of the reduced form.
- **Null space** $N(A)$ lives in R^n , dimension $n - r$. Solve $A\mathbf{x} = \mathbf{0}$.
- **Left null space** $N(A^T)$ lives in R^m , dimension $m - r$. Solve $\mathbf{y}^T A = \mathbf{0}$.

WHY THIS WORKS

Every matrix has four subspaces attached to it, and row reduction reveals all four at once. The rank r is the number of pivots, and it controls everything: two of the subspaces have dimension r , and the other two make up the difference. **The one thing to remember is that the basis for the column space comes from the ORIGINAL matrix, not the reduced one** — row operations change the column space, but they do not change *which* columns are pivot columns.

SOLUTION — EVERY STEP

1 Row-reduce, and notice the repeated row.

Row 3 is identical to row 1, so $R_3 - R_1$ wipes it out:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now clear above the second pivot with $R_1 - 2R_2$:

$$\rightarrow \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivots are in columns 1 and 2, so rank $r = 2$.

2 Column space — take the pivot columns of the ORIGINAL A .

$$\text{basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}, \quad \dim C(A) = 2$$

It is a 2-dimensional plane inside R^3 .

3 Row space — take the non-zero rows of the reduced form.

$$\text{basis} = \{ (1, 0, -2, 1), (0, 1, 1, 0) \}, \quad \dim C(A^T) = 2$$

This one lives in R^4 .

4 Null space — solve $Ax = 0$ from the reduced form.

The reduced rows say

$$x_1 - 2x_3 + x_4 = 0 \quad \text{and} \quad x_2 + x_3 = 0$$

Columns 3 and 4 have no pivot, so **x_3 and x_4 are free**. Express the others:

$$x_1 = 2x_3 - x_4, \quad x_2 = -x_3$$

Set one free variable to 1 and the other to 0, then swap:

$$\rightarrow \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \dim N(A) = 2$$

5 Left null space — find the combinations of rows that give zero.

You need y with $y^T A = 0$, that is $y_1(\text{row } 1) + y_2(\text{row } 2) + y_3(\text{row } 3) = 0$.

You already know one: row 3 = row 1, so row 1 - row 3 = **0**. That is $y = (1, 0, -1)$.

$$\text{basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}, \quad \dim N(A^T) = 1$$

6 Summarise in a table — this presentation earns marks.

Subspace	Lives in	Dimension	Basis
Column space $C(A)$	R^3	2	$(1,0,1), (2,1,2)$
Row space $C(A^T)$	R^4	2	$(1,0,-2,1), (0,1,1,0)$
Null space $N(A)$	R^4	2	$(2,-1,1,0), (-1,0,0,1)$
Left null space $N(A^T)$	R^3	1	$(1,0,-1)$

CHECK YOUR ANSWER

The two dimension identities must both hold.

- $\dim C(A^T) + \dim N(A) = 2 + 2 = 4 = \text{number of columns}$ ✓
- $\dim C(A) + \dim N(A^T) = 2 + 1 = 3 = \text{number of rows}$ ✓

Also verify a null-space vector directly: $A(2, -1, 1, 0) = (2 - 2 + 0 + 0, 0 - 1 + 1 + 0, 2 - 2 + 0 + 0) = (0, 0, 0)$ ✓

WATCH OUT

- **The column-space basis comes from the original A .** Using the reduced columns is the single most common mistake in this question. Row operations preserve which columns are pivots, but not the columns themselves.
- The null space and row space both live in R^4 ; the column space and left null space both live in R^3 . Getting these the wrong way round loses marks.
- The left null space is easy to skip. It is a quarter of the question — here it is one vector, found by spotting that row 3 equals row 1.

15. Complete solution of $A\mathbf{x} = \mathbf{b}$

Unit IV · Q7a · 6 marks · 15 minutes · overtime item

THE QUESTION

Find the complete solution of

$$x + 3y + 3z = 1$$

$$2x + 6y + 9z = 5$$

$$-x - 3y + 3z = 5$$

FORMULAS YOU NEED

Complete solution = one particular solution + everything in the null space:

$$\mathbf{x} = \mathbf{x}_p + c_1\mathbf{n}_1 + c_2\mathbf{n}_2 + \dots$$

- $\mathbf{x}_p$: set every free variable to 0 and solve.
- $\mathbf{n}_i$: solutions of $A\mathbf{x} = \mathbf{0}$, one for each free variable.

WHY THIS WORKS

If $\mathbf{x}_p$ solves $A\mathbf{x} = \mathbf{b}$ and $\mathbf{n}$ solves $A\mathbf{n} = \mathbf{0}$, then $A(\mathbf{x}_p + \mathbf{n}) = \mathbf{b} + \mathbf{0} = \mathbf{b}$, so the sum is also a solution.

Conversely the difference of any two solutions lies in the null space. So the solution set is one particular point with the whole null space slid onto it — a line, a plane, or a single point, depending on how many free variables there are.

SOLUTION — EVERY STEP

1 Write the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & 3 & 5 \end{array} \right]$$

2 Eliminate down the first column.

$$R_2 - 2R_1: (0, 0, 3 \mid 3)$$

$$R_3 + R_1: (0, 0, 6 \mid 6)$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 6 & 6 \end{array} \right]$$

3 Finish reducing.

$$R_3 - 2R_2: (0, 0, 0 \mid 0) \text{ — the third equation carried no new information.}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Pivots are in columns 1 and 3, so y is the free variable. The last row reads $0 = 0$, so the system is consistent and a solution exists.

4 Find a particular solution: set the free variable to zero.

Put $y = 0$. Row 2 gives $z = 1$. Row 1 gives $x + 0 + 3(1) = 1$, so $x = -2$.

$$\mathbf{x}_p = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

5 Find the null space: same reduction with zero on the right.

Row 2: $z = 0$. Row 1: $x + 3y + 0 = 0$, so $x = -3y$.

Take $y = 1$:

$$\mathbf{n} = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

6 Write the complete solution.

$$\mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \quad c \text{ any real number}$$

In coordinates: $x = -2 - 3c$, $y = c$, $z = 1$. **The solution set is a line in $\mathbb{R}^3$.**

CHECK YOUR ANSWER

Substitute the general solution back into all three equations, keeping c symbolic. If it works for every c , you have the whole family, not just one point.

- $(-2-3c) + 3c + 3(1) = -2 + 3 = 1 \quad \checkmark$
- $2(-2-3c) + 6c + 9(1) = -4 - 6c + 6c + 9 = 5 \quad \checkmark$
- $-(-2-3c) - 3c + 3(1) = 2 + 3c - 3c + 3 = 5 \quad \checkmark$

The c terms cancel in every equation — that is the signature of a correct null-space vector.

WATCH OUT

- **Check consistency first.** If a row reduces to $(0 \ 0 \ 0 \mid k)$ with $k \neq 0$, there is **no solution** and you should say so rather than forcing an answer.
- Free variables are the ones whose columns have **no pivot**. Here it is y , not z — read it from the reduced matrix, do not assume it is the last variable.
- Present the answer in the form $\mathbf{x}_p + c\mathbf{n}$. Giving only $\mathbf{x}_p$ answers a different question and loses half the marks.

16. Composition of matrix transformations

Unit III · Q6a · 6 marks · 12 minutes · overtime item

THE QUESTION

Determine the matrix that describes a reflection in the x -axis, followed by a rotation through $\pi/2$, followed by a contraction of factor $1/3$. Find the image of the point $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ under this sequence of mappings.

FORMULAS YOU NEED

- Reflection in the x -axis: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (y changes sign)
- Reflection in the y -axis: $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (x changes sign)
- Rotation through θ : $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
- Dilation or contraction by factor k : kI — just multiply the whole matrix by k

WHY THIS WORKS

Applying transformation S first and then T means computing $T(S(\mathbf{x}))$, which in matrices is $T(S\mathbf{x}) = (TS)\mathbf{x}$. So the matrix of the composition is the product with **the first transformation on the right**. Matrix multiplication is not commutative, so the order genuinely matters: reflect-then-rotate is a different map from rotate-then-reflect. **Read the question left to right and build the product right to left.**

SOLUTION — EVERY STEP

- 1 Write down the three matrices in the order the question gives them.**

$$\text{Reflect in } x: R_f = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\text{Rotate } \pi/2: \cos(\pi/2) = 0, \sin(\pi/2) = 1, \text{ so } R_\theta = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Contract by } 1/3: C = \frac{1}{3} I$$

- 2 Assemble the product in reverse order.**

$$M = C \cdot R_\theta \cdot R_f = \frac{1}{3} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The reflection is applied first, so it sits on the right.

3 Multiply the two matrices.

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Row 1 $\times$ column 1: $(0)(1) + (-1)(0) = 0$

Row 1 $\times$ column 2: $(0)(0) + (-1)(-1) = 1$

Row 2 $\times$ column 1: $(1)(1) + (0)(0) = 1$

Row 2 $\times$ column 2: $(1)(0) + (0)(-1) = 0$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

4 Include the contraction factor.

$$M = \frac{1}{3} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/3 \\ 1/3 & 0 \end{bmatrix}$$

5 Apply M to the point.

$$M \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 4/3 \end{bmatrix}$$

The image is $(1/3, 4/3)$.

CHECK YOUR ANSWER

Track the point through the three stages by hand — if it matches, your matrix order is right.

- Start: $(4, 1)$
- Reflect in the x -axis (flip the sign of y): $(4, -1)$
- Rotate $\pi/2$ (the rule $(x, y) \rightarrow (-y, x)$): $(1, 4)$
- Contract by $1/3$: **$(1/3, 4/3)$** ✓

This matches the matrix answer, so the order $C R_\theta R_f$ is correct.

WATCH OUT

- **Order is everything.** The first transformation mentioned goes on the **right** of the product. Building it left to right gives a different matrix and no marks.
- Reflection in the x -axis flips **y** , giving $\text{diag}(1, -1)$. It is easy to write $\text{diag}(-1, 1)$ by mistake — that is reflection in the y -axis.
- A **contraction** has $k < 1$ and a **dilation** has $k > 1$, but the matrix is kI either way. Do not invert the factor.
- Always do the by-hand check above. It takes twenty seconds and catches the order mistake.